

Empirical Analysis of Multi-Floor 3D Scene Generation using Open Weight Models

Sowndaryan Jayaprakash Anand, Neeresh Perla Kumar, Ming Shao
Miner School of Computer and Information Sciences
University of Massachusetts Lowell
Lowell, MA, USA

Abstract—Physically realistic and interactive 3D simulation environments are important for evaluating navigation and long-horizon task planning in embodied artificial intelligence (EAI). However, existing multi-floor scene-generation pipelines often rely on closed-source cloud models, introducing inference costs and reproducibility challenges. We present the Open-Weight Pipeline (OWP), a fully localized adaptation of the MANSION multi-floor scene-generation pipeline using a locally deployed open-weight model. OWP enables end-to-end multi-floor scene generation through controlled decoding and structured output generation without cloud API calls. We evaluate OWP against MANSION on hospital and residential scenes and across additional indoor typologies. OWP achieves comparable navigability, with 100% reachability in the hospital scene and 97.3% in the residential scene. It also reduces normalized out-of-boundary violations by 25.0% and 12.1% in the hospital and residential scenes, respectively, while reducing the normalized collision rate by 39.5% in the residential evaluation. These results demonstrate the feasibility of open-weight models for reproducible and cost-efficient multi-floor 3D scene generation.

I. INTRODUCTION

Embodied artificial intelligence (EAI) and robotic navigation rely on physically realistic and interactive 3D simulation environments [1] for training, evaluation, and benchmarking of autonomous agents. Procedural scene generation [2] provides a scalable alternative to manually constructing such environments by automatically generating diverse indoor layouts, objects, and spatial configurations. Recent language-guided scene-generation frameworks, such as MANSION [3], use large language models (LLMs) as high-level spatial planners to translate natural-language specifications into structured architectural representations, including multi-room and multi-floor layouts. MANSION [1] extends procedural 3D scene generation to multi-story environments through a multi-stage LLM-driven pipeline. However, its reliance on closed-source cloud models introduces three practical limitations: recurring API costs, limited flexibility for multi-model experimentation, and restricted control over inference parameters needed for reliable structured generation. Local open-weight models [4] address these limitations by enabling configurable and reproducible inference without per-request API charges.

II. BACKGROUND

A. MANSION Overview

The baseline MANSION pipeline decomposes language-to-3D building generation into a hierarchical, multi-stage work-

flow coordinated by specialized LLM reasoning agents, as illustrated in Fig. 1. Starting from a high-level natural-language description, the pipeline progressively generates building-level constraints, floor-specific spatial programs, geometric floorplans, and finally a populated 3D simulation environment.

1) Whole-Building Planning: Given a high-level prompt, such as “Get me a small three-story office building,” the Chief Designer establishes global architectural constraints for the complete structure. These include the overall architectural style, surface and material guidelines, vertical transitions such as staircases and elevators, and the functional zoning assigned to each floor. This stage provides the global structure that constrains subsequent floor-level planning.

2) Per-Floor Planning: For each floor ($F_1, F_2, \dots, F_N$), an independent Floor Designer generates a localized spatial program. Each floor is represented using structured Room ID Cards and a bubble diagram. Room ID Cards specify properties such as room identity, functional type, target area, material style, wall texture, and adjacency requirements. The bubble diagram provides a topological representation in which nodes correspond to rooms and edges represent connectivity relationships between them.

3) Floorplan Synthesis: The room specifications and bubble diagrams are then converted into continuous 2D floorplans by a geometric solver. This stage determines room placement, bounding regions, wall partitions, and the alignment of shared vertical structures across floors. The resulting floorplans provide the geometric representation required for downstream 3D scene construction.

This stage determines room placement, bounding regions, wall partitions, and cross-floor vertical alignment, producing the geometric floorplans required for downstream 3D scene construction.

4) Scene Instantiation: Finally, the generated floorplans are transformed into physically interactive 3D environments in AI2-THOR. This stage includes material assignment, door placement, and retrieval and placement of furniture and other scene objects according to room function and spatial constraints. The resulting environment can then be used for embodied-agent navigation and interaction tasks.

III. LOCALIZED FRAMEWORK

In the baseline MANSION framework, Stages I, II, and IV rely on sequential calls to cloud-based LLM APIs. Because

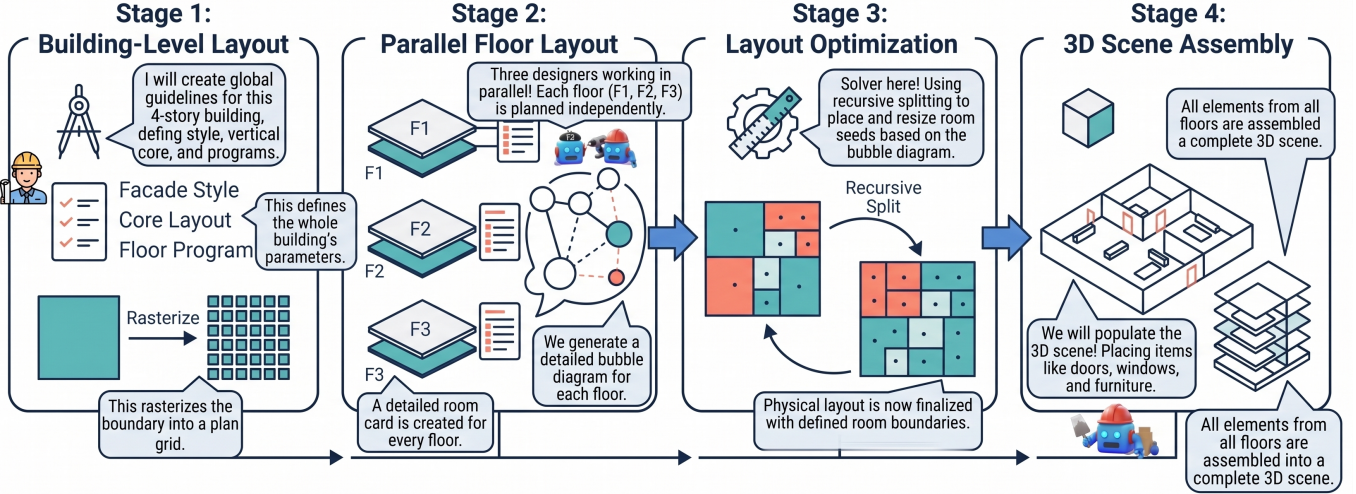


Fig. 1: Overview of the MANSION generation pipeline.

the planning stages generate extensive JSON-formatted Room ID Cards, bubble diagrams, and scene specifications, multi-floor generation requires many consecutive model calls. Our localized framework replaces these cloud endpoints with an inference-optimized local pipeline, as illustrated in Figure 2.

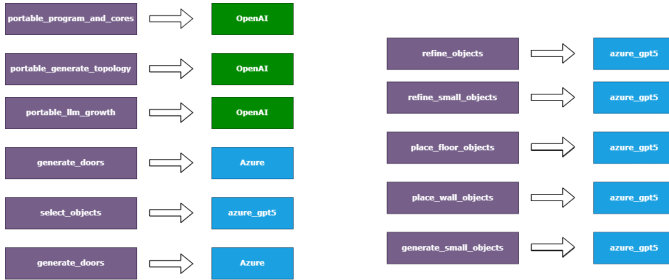


Fig. 2: Overview of API calls in MANSION pipeline

Our framework replaces the cloud-based model endpoints with a locally deployed open-weight model, Gemma4:12B [8], running through Ollama [10]. The evaluation system uses an Intel Core i5-12600K CPU, an NVIDIA GeForce RTX 4080 GPU, 32, GiB of system memory, and 1, TB of local storage. This configuration was selected to evaluate whether multi-floor scene generation could be performed using a single local workstation rather than external cloud inference services.

Initial experiments with the local model exhibited output truncation and JSON parsing failures, particularly when generating large structured specifications containing multiple rooms, spatial relationships, and object-placement information. These failures prevented later stages of the pipeline from correctly parsing and processing the generated outputs. To improve reliability, we evaluated different context-window sizes, output-token limits, and sampling parameters.

The final configuration uses a context window $num_context$ of 32,768 tokens and an output limit ($num_predict$) of 8,192 tokens. The larger context window

allows the model to retain long architectural prompts and intermediate scene specifications, while the increased output limit provides sufficient capacity to generate complete structured responses. In our evaluated scenes, these settings prevented the truncation failures observed during initial experiments.

We further applied low-temperature decoding ($T = 0.1$, $top_p = 0.9$) to reduce generation variability and improve consistency across repeated structured outputs. This was particularly important because downstream stages expect well-formed JSON structures with predefined fields and spatial relationships. Combined with schema-constrained generation [9], these settings improved the reliability of structured outputs across the procedural planning stages.

With the optimized inference configuration, the localized pipeline successfully executed the complete multi-floor generation workflow end-to-end on local hardware without cloud API calls or manual intervention. This setup also provides direct control over inference parameters, making it easier to modify and evaluate different open-weight models within individual stages of the pipeline.

IV. EXPERIMENTS

A. Experiment Setup

We evaluate our open-weight pipeline (OWP) through a direct comparative case study against the baseline MANSION framework under identical experimental conditions ($N=2$). To ensure an exact and reproducible comparison, all natural language specifications and prompt templates are aligned with the benchmark metadata from the MansionWorld [7] dataset, which serves as the source for all comparative building scenes. Specifically, our input prompts replicate the dataset’s architectural conditions by including both the target building typology and explicit total area limits (e.g., “*small flat residential studio, area: 300*”). We benchmark across two representative multi-level architectural typologies: a Hospital Facility and

TABLE I: Architectural Layout and Room Breakdown

Metric	MANSION	OWP (Ours)
Rooms	16 (8/8)	11 (6/5)
Doors	17 (9/8)	11 (6/5)
Windows	31 (15/16)	33 (15/18)
Beds	3 (0/3)	4 (1/3)
Evaluated Rooms	18 (9/9)	5 (4/1)

Values in parentheses indicate F1/F2 counts.

TABLE II: Asset Population and Physical Quality Metrics

Metric	MANSION	OWP (Ours)
Objects (Small)	341 (47)	64 (3)
Out-of-Bound (#OB)	135	19
Collisions (#CN)	101	18
Reachability (#Rch)	94.8%	100%

a Multi-Story Residential duplex. We also provide various different standalone scenes generated from OWP. Evaluation is categorized across three axes: visual and geometric fidelity, structural navigability, and physical collision adherence.

B. Hospital Scene Generation Results

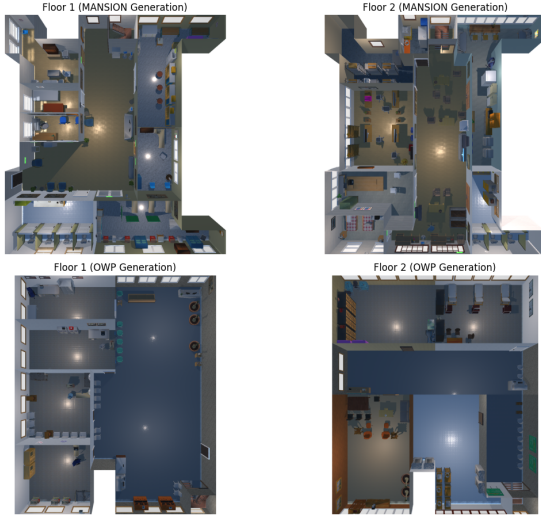


Fig. 3: Qualitative comparison of a 2-story hospital facility. The top row illustrates MANSION outputs; bottom row illustrates OWP outputs.

As observed in the generated floorplans (Figure 3), OWP synthesizes coherent, topologically valid multi-floor layouts alongside the closed-weight MANSION framework. Room partition zoning and primary structural furniture anchors remain consistent across both floors.

C. Empirical Analysis of Hospital Scene Benchmark

The quantitative benchmarks presented in Table I and Table II highlight a distinct trade-off between semantic object density [11] and physical geometric validity [12] when deploying OWP relative to cloud baselines:

Navigability and Reachability: OWP achieves an Average Reachability (#Rch) of 100%, compared with 94.8% for the MANSION baseline. This complete coverage indicates that all evaluated functional regions in the OWP scene remain reachable under the navigation evaluation. The result suggests that the localized pipeline preserves the topological connectivity required for downstream embodied-agent navigation despite differences in generated scene complexity.

Boundary Confinement and Physical Grounding: OWP substantially reduces the absolute number of geometric violations relative to MANSION. Out-of-Boundary Violations (#OB) decrease from 135 to 19, corresponding to an 85.9% reduction, while Object Collisions and Overlaps (#CN) decrease from 101 to 18, corresponding to an 82.2% reduction. However, these absolute counts are influenced by the substantially different asset populations: MANSION generates 341 objects, whereas OWP generates 64.

Normalized Physical Quality: Normalizing geometric violations by the number of generated objects provides a more direct comparison of physical validity independent of overall asset population. The normalized Out-of-Boundary rate is calculated as

$$R_{OB} = \frac{\#OB}{N_{objects}} \times 100,$$

while the normalized Collision rate is calculated as

$$R_{CN} = \frac{\#CN}{N_{objects}} \times 100.$$

For MANSION, the normalized Out-of-Boundary rate is $R_{OB}^{MANSION} = \frac{135}{341} \times 100 = 39.59$, while OWP produces

$$R_{OB}^{OWP} = \frac{19}{64} \times 100 = 29.69.$$

This represents a 9.90-point reduction in the normalized Out-of-Boundary rate, equivalent to a 25.0% relative reduction with respect to the MANSION rate. For Collision events, MANSION produces

$$R_{CN}^{MANSION} = \frac{101}{341} \times 100 = 29.62,$$

compared with

$$R_{CN}^{OWP} = \frac{18}{64} \times 100 = 28.13$$

for OWP. This corresponds to an approximately 5.0% lower collision rate per 100 instantiated objects for OWP.

These normalized results provide a more clear interpretation than the raw violation counts alone. While OWP's substantially lower object population explains a portion of the reduction in total collision and boundary violations, the localized pipeline also shows a lower rate of Out-of-Boundary Violations per instantiated object and a comparable Collision rate. Therefore, the results support improved boundary confinement without attributing the entire reduction in physical violations only to the lower asset density of OWP.

Asset Density and Scaling Trade-Off: MANSION populates the evaluated scene with substantially more objects than OWP, with 341 objects across 18 evaluated rooms compared with 64 objects across 5 evaluated rooms. This difference highlights an important trade-off in the localized pipeline. OWP’s decoding configuration ($T=0.1$) favors structural stability and valid JSON generation, but also results in fewer small and medium assets. However, the lower absolute number of collisions and boundary violations should not be interpreted as evidence of superior geometric optimization. Instead, the normalized results indicate that OWP maintains comparable object-level collision behavior while providing a substantially lower rate of boundary violations per instantiated object.

D. Multi-Story Residential Generation Results

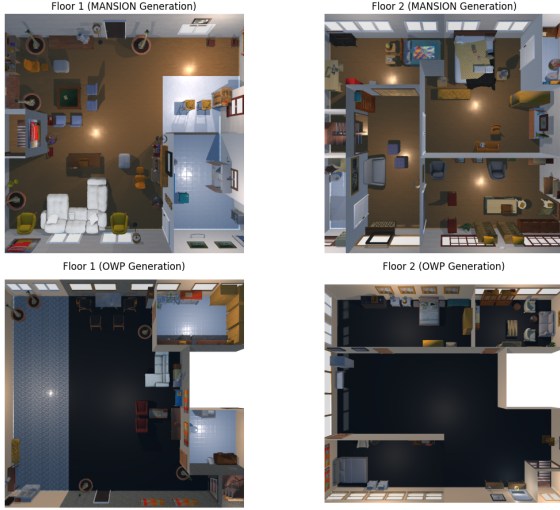


Fig. 4: Qualitative comparison of a 2-story residential townhouse

E. Empirical Analysis of Multi-Story Residential

Object Density and Scaling: OWP instantiates 149 total objects across 10 rooms (compared to MANSION’s 207 objects across 11 rooms), without suffering generation failures as illustrated in Table III. The lower absolute number of collisions and boundary violations should not be interpreted as evidence of superior geometric optimization. Instead, the normalized results indicate that OWP maintains comparable object-level collision behavior while providing a substantially lower rate of boundary violations per instantiated object.

Navigability and Reachability: When benchmarked against MANSION’s closed-weight pipeline, OWP achieves an Average Reachability ($\#Rch$) of 97.3%, trailing the MANSION baseline (98.6%) by only 1.3%. This narrow margin suggests that open-weight models, when constrained by structured JSON schemas and deterministic decoding parameters, can perform similarly to closed-source models in spatial adjacency reasoning, door placement, and agent navigation path clearance.

TABLE III: Architectural Layout and Room Breakdown (Residential Scene)

Metric	MANSION	OWP (Ours)
Rooms	11 (5/6)	10 (5/5)
Doors	10 (4/6)	8 (4/4)
Windows	18 (6/12)	26 (10/16)
Beds	2 (0/2)	2 (0/2)
Evaluated Rooms	12 (5/7)	10 (5/5)

Values in parentheses indicate F1/F2 counts.

TABLE IV: Asset Population and Physical Quality Metrics (Residential Scene)

Metric	MANSION	OWP (Ours)
Objects (Small)	207 (10)	149 (0)
Out-of-Bound ($\#OB$)	79	50
Collisions ($\#CN$)	179	78
Reachability ($\#Rch$)	98.6%	97.3%

Boundary Confinement and Physical Grounding: Where OWP performs differently is in grounding and spatial precision: OWP incurs only 50 out-of-boundary violations ($\#OB$) compared to MANSION’s 79, representing a 36.7% reduction in structural wall-clipping and room perimeter penetrations. Inter-object overlaps drop from 179 ($\#CN$) in MANSION to 78 in our framework, achieving a 56.4% decrease in overlapping bounding volumes.

Applying the same object-normalization procedure used for the hospital scene described in Section IV-C, the residential results are summarized in Table V. OWP reduces the normalized Out-of-Boundary rate from 38.16 to 33.56 violations per 100 objects, corresponding to a 12.1% relative reduction. The normalized Collision rate decreases from 86.47 to 52.35 collisions per 100 objects, corresponding to a 39.5% relative reduction. These results indicate that the reduction in physical violations observed for OWP is not solely attributable to its lower overall object count, with the largest improvement occurring in collision frequency.

V. STANDALONE RESULTS

Beyond the MANSION comparisons, we evaluate OWP across diverse indoor typologies, with representative generations shown in Figure 5 and quantitative results summarized in Table VI. OWP generates across residential, school, townhouse, and duplex scenes while maintaining generally high reachability and constrained physical violations. The *School* scene generated 205 objects across 9 evaluated rooms with 100.0% Average Reachability ($\#Rch$) and 31 collision events ($\#CN$). While *School* and *Duplex* achieved 100.0% reachability, the compact multi-story *Townhouse* scene reached only 76.4%, indicating that narrow floor envelopes and stair transitions remain a key constraint for conservative object placement.



Fig. 5: Samples from four different scenes generated by OWP.

TABLE V: Object-Normalized Physical Quality Metrics for Residential Scene

Metric	MANSION	OWP (Ours)
OB / 100 Objects	38.16	33.56
CN / 100 Objects	86.47	52.35

TABLE VI: Asset Population and Physical Quality Metrics Across Typologies

Typology	Rooms	Objects	OB	CN
Residential	10 (5/5)	98 (0)	37	41
School	9 (4/5)	205 (5)	64	31
Townhouse	9 (4/5)	98 (1)	35	24
Duplex	4 (4/0)	71 (1)	28	40

A. Limitations and Future Work

While low-temperature inference improves geometric validity, it can reduce micro-asset density and clutter diversity and may restrict agent traversal in compact layouts. This trade-off is particularly evident in narrow multi-story environments, where conservative object placement can reduce navigable space. Future work will evaluate larger open-weight models to determine whether increased model capacity improves asset placement while preserving structured generation reliability. We will also explore heterogeneous pipeline configurations that use specialized models for different stages, such as architectural zoning, floorplan reasoning, and object placement.

VI. CONCLUSION

In this paper, we presented the Open-Weight Pipeline (OWP), a fully localized framework for multi-floor 3D procedural scene generation that replaces cloud-based model inference with locally deployed open-weight models. By combining expanded context and output limits, low-temperature decoding, and schema-constrained structured generation, OWP enables the complete multi-stage scene-generation workflow to run end-to-end on local hardware without cloud API calls or manual intervention.

Typology Reachability (%)

Typology	Reachability (%)
Residential	96.0
School	100.0
Townhouse	76.4
Duplex	100.0

Room values in parentheses indicate F1/F2 counts; object values in parentheses indicate small objects.

Our experiments show that OWP achieves navigational performance comparable to the MANSION baseline while improving several measures of physical scene quality. Across the hospital and residential evaluations, OWP achieved up to 100% reachability, reduced normalized out-of-boundary violation rates by up to 25.0%, and reduced normalized collision rates by up to 39.5%. Additional evaluations across residential, school, townhouse, and duplex scenes further demonstrate that the pipeline can generate diverse multi-floor indoor environments, although performance decreases in compact layouts with constrained navigation paths. Overall, these results demonstrate the feasibility of open-weight models as a reproducible and cost-efficient alternative for multi-floor 3D scene generation.

REFERENCES

- [1] L. Che, S. Wen, S. Huang, C. Wang, Y. Yang, G. Dudek, X. Wang, and J. Su, “MANSION: Multi-floor lANguage-to-3D Scene generatIOn for loNg-horizon tasks,” 2026, *arXiv:2603.11554*. Available: <https://arxiv.org/abs/2603.11554>
- [2] X. Liang, M. Tao, Y. Xia, T. Shi, J. Wang, and J. Yang, “Self-evolving Agents with reflective and memory-augmented abilities,” 2024, *arXiv:2409.00872*. Available: <https://arxiv.org/abs/2409.00872>
- [3] Y. Ma, Z. Song, Y. Zhuang, J. Hao, and I. King, “A Survey on Vision-Language-Action Models for Embodied AI,” *IEEE Transactions on Neural Networks and Learning Systems*, 2024, doi: 10.48550/arXiv.2405.14093.
- [4] H. Fu, R. Jia, L. Gao, M. Gong, B. Zhao, S. Maybank, and D. Tao, “3D-FRONT: 3D furnished rooms with layOuts and semaNTics,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (ICCV)*, Oct. 2021, pp. 10933–10942, doi: 10.48550/arXiv.2011.09127
- [5] B. T. Willard and R. Louf, “Efficient guided generation for large language models,” 2023, *arXiv:2307.09702*. [Online]. Available: <https://arxiv.org/abs/2307.09702>
- [6] E. Kolve, R. Mottaghi, W. Han, E. VanderBilt, L. Weihs, A. Herrasti, D. Gordon, Y. Zhu, A. Gupta, and A. Farhadi, “AI2-THOR: An interactive 3D environment for visual AI,” 2017, *arXiv:1712.05474*. Available: <https://arxiv.org/abs/1712.05474>
- [7] superbigsaw, “MansionWorld,” *Hugging Face Datasets*, 2026. [Online]. Available: <https://huggingface.co/datasets/superbigsaw/MansionWorld>.
- [8] Gemma Team, S. El Abd, V. Aggarwal, R. Algayres *et al.*, “Gemma 4 Technical Report,” 2026, *arXiv:2607.02770*. Available: <https://arxiv.org/abs/2607.02770>
- [9] S. Lundberg, M. Ribeiro, and Guidance Contributors, “Guidance: A language for controlling large language models,” 2024. [Online]. Available: <https://github.com/guidance-ai/guidance>
- [10] Ollama Authors, “Ollama: Get up and running with large language models locally,” 2023. [Online]. Available: <https://github.com/ollama/ollama>
- [11] Y. Yang, M. Schwonberg, K. Ehsani, E. VanderBilt, A. Gupta, R. Mottaghi, and L. Weihs, “Holodeck: Language-guided generation of 3D embodied AI environments,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16223–16233, doi: 10.48550/arXiv.2312.09067.
- [12] G. Shi, S. Huang, S. Song, and R. Gao, “PhyScene: Physically Plausible 3D Scene Generation with Embodied Physical Simulation,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024, doi: 10.48550/arXiv.2404.09455.